

Parallel lines and the covariant derivative

7

We never remark any passion or principle in others, of which, in some degree or other, we may not find a parallel in ourselves.

David Hume (1711–1776) *A Treatise of Human Nature*

Comparisons are odorous

William Shakespeare (1564–1616)

Much Ado About Nothing III:5

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In order to describe physical quantities in general relativity, we shall need to evaluate mathematical objects (functions, vector, and tensor fields and so forth) at particular points in spacetime. We shall also identify differential equations for these objects to understand how they change with position in spacetime. This requires the notion of a derivative. When spacetime is curved, some of our basic assumptions about vectors and their derivatives break down. This chapter is concerned with finding a method to take derivatives of vectors with respect to position in spacetime in cases where the spacetime is curved.

Example 7.1

Probably the most familiar example of a curved space is the one in which we live: the Earth's surface. In navigating around our home town, we might use a street map, the coordinates for which are based on a two-dimensional rectangular grid based on north-south and east-west axes. But this street map only works locally and can't be extended to the whole planet because of the curvature of the Earth. Nevertheless, we can imagine smoothly transitioning between lots of small rectangular maps to cover the whole surface of the globe.

In much the same way, (3+1)-dimensional spacetime can be covered smoothly using lots of locally flat maps based on four coordinates. A spacetime that can be covered by a smoothly changing set of coordinates is known in mathematics as a **manifold**.¹ The study of smoothly changing spaces is known as **differential geometry** and is the subject of Part V of this book. Owing to its smoothness, a manifold describing spacetime is necessarily **flat** over a sufficiently small region, across which it looks identical to the Minkowski spacetime encountered in the first part of this book. Over larger distances however, the spacetime might be curved, and it is this curvature that we describe over the next few chapters, with the derivative formulated in this chapter an essential first step.

¹We can think of a two-dimensional manifold as being made of rubber. It can be a sphere or a torus, but the crucial thing about it is that it shouldn't be punctured in any way: it should be smooth and continuous. Smoothness is a property we'll return to regularly.

²For example, if we think of the surface of the Earth then a straight-line path from New York to Paris would plough through the Earth, burrowing hundreds of kilometres underground. This directed straight-line path is then outside the space we are trying to describe.

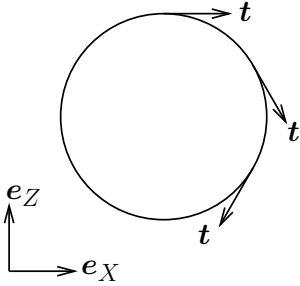


Fig. 7.1 A tangent vector $\mathbf{t}$ is parallel transported around a surface. Its components are always the same in the local coordinate system, but $\mathbf{t}$ changes when viewed in the (X, Z) coordinate system set up by observers able to embed the space in higher dimensions.

³Imagine a tourist using a street map in London. In absolute terms, their North-direction is rather different to the North-direction of an analogous tourist in Tokyo, even though it is analogously defined in both city street maps as a particular vector tangent to the Earth. The North-direction could be thought of as being parallel transported between the two cities.

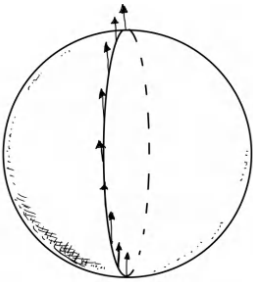


Fig. 7.2 Failure of parallelism for sphere embedded in $\mathbb{R}^3$. The vectors drawn on the surface of the sphere all point in the same absolute direction, according to the embedding in $\mathbb{R}^3$, but they only lie in the tangent plane at one point; more often, they are pointing out of the tangent plane.

7.1 Parallelism

Consider a curved spacetime where observers are confined. The picture to have in mind is of ants confined to a two-dimensional surface such as a football, or of humans confined to the surface of the earth. All measurements are to be made in the surface, so the observers are not allowed to float above it to take advantage of its being embedded in three-dimensional space. The idea of a vector as a directed straight line joining two points is fine for flat space, but ceases to be of much use in a curved space.² The notion of a vector as a **tangent** to a path is, however, of much more use. Picture how the **tangent vector** to a path on a two-dimensional surface embedded in three-dimensional space will change its direction as we move it around the curved space, in order for it to still lie in the tangent plane of the surface. This behaviour of the tangent vector will provide a measure of vectors being **parallel**.

Next, we imagine that the path in the surface is one that doesn't change direction according to the observers (e.g. a great circle on a sphere, as shown in Fig. 7.1). The tangent vector of this path should, according to the trapped ants, be 'the same', or parallel, at all points on the path. A tangent vector to the path at some point may then be transported to a different point on the path and, if it is identical to the tangent vector determined at its new position, then we say that the vector has been **parallel transported** (Fig. 7.1). From the point of view of the observers confined to the surface, two vectors can then be compared at two different points in spacetime.

To generalize beyond tangent vectors: our trapped ants set up a coordinate system with which to make measurements. Measurements are always made locally, so they parallel transport their set of axes to the point where they want to measure the orientation of a vector. (The local coordinate system might, for example, use the tangent vector described above, and another axis in the surface orthogonal to this direction.) From this point of view, any vector that has the same components in each of the local coordinate systems is judged to be parallel at the different points. However, these vectors *appear* to change directions when we view the surface as embedded in a higher dimensional space, just as the coordinate system used by the ants on the surfaces appears to change with position (Fig. 7.1).³ This concept of parallelism will be included when we describe derivatives in curved spacetime, since the derivative of a field of parallel vectors should come out to be zero.

Example 7.2

The vectors in Fig. 7.2 are all parallel in three-dimensional space $\mathbb{R}^3$. However, for observers living on the surface of the sphere, the vectors are not parallel: they all make different angles to the tangent plane of the sphere's surface. The result of correctly parallel transporting a vector along several paths on a spherical surface is shown in Fig. 7.3. The components of the vector are identical in each of the local coordinate systems that the confined observers set up.

7.2 Derivatives and connections

We now turn to a method to evaluate the change in a vector with position. In order to evaluate, via a derivative, the change in a vector as it is transported around, we need to disentangle two effects. The first is the intrinsic change in the vector with position, which is what we want the derivative to output. The second is the change in the vector reflecting the fact that the coordinates (or, equivalently, the basis vectors) change in different parts of space. To extract the intrinsic change we define a new kind of derivative of the vector. This is the **covariant derivative** which evaluates the intrinsic change in the vector $\mathbf{v}$. We can make sense of the covariant derivative conceptually as

$$\left(\begin{array}{c} \text{Covariant} \\ \text{derivative of } \mathbf{v} \end{array} \right)_u = \left(\begin{array}{c} \text{Change in} \\ \text{vector } \mathbf{v} \end{array} \right) - \left(\begin{array}{c} \text{Change due to} \\ \text{coordinate system} \end{array} \right). \quad (7.1)$$

This derivative is **directional**: it tells us the change in $\mathbf{v}$ as we move along the vector $\mathbf{u}$ (hence the subscript in the previous equation).

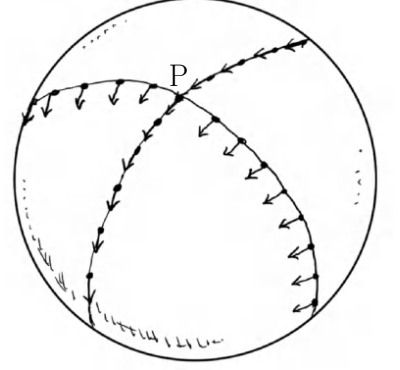


Fig. 7.3 Parallel transport of a vector on a spherical surface.

Example 7.3

The covariant derivative in a curved space generalizes the notion of a directional derivative in ordinary calculus. The gradient of a surface of constant f in Euclidean 3-space is given by

$$\vec{\nabla} f(x, y, z) = \frac{\partial f}{\partial x} \vec{e}_x + \frac{\partial f}{\partial y} \vec{e}_y + \frac{\partial f}{\partial z} \vec{e}_z. \quad (7.2)$$

This is interpreted as a vector normal to the tangent plane of the surface of constant f . If we want to know the change of $f(x, y, z)$ along a particular vector $\vec{u}$ we use the directional derivative, defined as

$$\vec{u} \cdot \vec{\nabla} f = u^x \frac{\partial f}{\partial x} + u^y \frac{\partial f}{\partial y} + u^z \frac{\partial f}{\partial z}. \quad (7.3)$$

This can be thought of as

$$\vec{u} \cdot \vec{\nabla} f = \left(\begin{array}{c} \text{Value of } f \\ \text{at tip of } \vec{u} \end{array} \right) - \left(\begin{array}{c} \text{Value of } f \\ \text{at base of } \vec{u} \end{array} \right). \quad (7.4)$$

Now to take the derivative. Consider the vector $\mathbf{v} = v^\mu \mathbf{e}_\mu$. We take a derivative with respect to the coordinates, allowing both the components v^μ and the basis vectors $\mathbf{e}_\mu$ to change in spacetime. The derivative we shall take is $\partial/\partial x^\alpha$, which should be thought of as the directional derivative along the direction $\mathbf{e}_\alpha$. Employing the Leibniz product rule, we have

$$\frac{\partial \mathbf{v}}{\partial x^\alpha} = \frac{\partial v^\mu}{\partial x^\alpha} \mathbf{e}_\mu + v^\mu \frac{\partial \mathbf{e}_\mu}{\partial x^\alpha}. \quad (7.5)$$

The tricky second term on the right is due to the change in basis vectors with position. The derivative of the basis vector $\mathbf{e}_\mu$ can have components along any of the basis vectors, so to express this we define **connection coefficients**, also known as **Christoffel symbols**,⁴ denoted $\Gamma^\mu_{\alpha\beta}$, and

⁴Elwin Bruno Christoffel (1829–1900). The mathematics described here were invented by Christoffel in 1869 and further explored by Gregorio Ricci-Curbastro (1853–1925) who, in the years leading to 1900, developed the mathematical machinery employed by Einstein in developing general relativity. We shall follow the modern convention of calling the symbols ‘connection coefficients’ in this book.

then write the change in basis vectors as

$$\frac{\partial \mathbf{e}_\mu}{\partial x^\alpha} = \Gamma^\lambda_{\alpha\mu} \mathbf{e}_\lambda. \quad (7.6)$$

The connection coefficients encode all of the information describing how the coordinates change as we move around spacetime. Another way of thinking about this is that, since there are different local coordinate systems at different points in space, the connection coefficients tell us how the coordinate systems are *connected*, that is, how to translate between coordinate system as we move around.⁵

⁵Over the next few chapters, we shall see how the connection coefficients, whose values are functions of position of spacetime, can be used to tell us about the curvature of spacetime itself. They can be derived from the metric, which is ultimately the source of curvature.

⁶Recall that

$$\mathbf{e}_r = \left(\frac{\partial x}{\partial r} \right)_\theta \mathbf{e}_x + \left(\frac{\partial y}{\partial r} \right)_\theta \mathbf{e}_y,$$

which yields, from $x = r \cos \theta$ and $y = r \sin \theta$, the expression

$$\mathbf{e}_r = \cos \theta \mathbf{e}_x + \sin \theta \mathbf{e}_y.$$

Similarly

$$\mathbf{e}_\theta = \left(\frac{\partial x}{\partial \theta} \right)_r \mathbf{e}_x + \left(\frac{\partial y}{\partial \theta} \right)_r \mathbf{e}_y,$$

or

$$\mathbf{e}_\theta = -r \sin \theta \mathbf{e}_x + r \cos \theta \mathbf{e}_y.$$

⁷We shall see in Chapter 11 that a combination of the coefficients and their derivatives is needed to tell us whether the space is curved or not.

⁸In summary: $\Gamma^\lambda_{\alpha\mu}$ is the rate at which the vector $\mathbf{e}_\mu$ rotates towards $\mathbf{e}_\lambda$ as we travel along a curve whose tangent vector is $\mathbf{e}_\alpha$. Notice that we choose the order of the downstairs indices in $\Gamma^\lambda_{\alpha\mu}$ to start with the component with respect to which we take the derivative in eqn 7.6. This convention is not universal: Hawking and Ellis use it; Misner, Thorne, and Wheeler do not. Since $\Gamma^\lambda_{\alpha\mu} = \Gamma^\lambda_{\mu\alpha}$ in all of the coordinate frames we shall deal with, this is not a crucial point.

Example 7.4

In the following two chapters, we shall find a simple and efficient means of extracting connection coefficients. Before we get to that, here is a simple, ‘brute force and ignorance’ example, based on eqn 7.6. We saw in Chapter 3 that for a plane-polar coordinate system, the basis vectors have derivatives⁶

$$\begin{aligned} \frac{\partial \mathbf{e}_r}{\partial r} &= 0, & \frac{\partial \mathbf{e}_r}{\partial \theta} &= \frac{\mathbf{e}_\theta}{r}, \\ \frac{\partial \mathbf{e}_\theta}{\partial r} &= -\frac{\mathbf{e}_r}{r}, & \frac{\partial \mathbf{e}_\theta}{\partial \theta} &= -r \mathbf{e}_r. \end{aligned} \quad (7.7)$$

This allows us to write down the connection coefficients. Using eqn 7.6 we can read off

$$\begin{aligned} \Gamma^r_{rr} &= 0, & \Gamma^\theta_{rr} &= 0, & \Gamma^r_{\theta r} &= 0, & \Gamma^\theta_{\theta r} &= \frac{1}{r}, \\ \Gamma^r_{r\theta} &= 0, & \Gamma^\theta_{r\theta} &= \frac{1}{r}, & \Gamma^r_{\theta\theta} &= -r, & \Gamma^\theta_{\theta\theta} &= 0. \end{aligned} \quad (7.8)$$

Although Euclidean space described by the plane polar coordinates is flat, we still have non-zero connection coefficients. The presence of connection coefficients therefore does not alone tell us whether a space is curved.⁷

In all of the coordinate frames that we examine in this book, the connection coefficients have the property

$$\Gamma^\mu_{\alpha\beta} = \Gamma^\mu_{\beta\alpha}. \quad (7.9)$$

A connection with this property is often called symmetric or **torsion free**.

Finally, we ask why we call the Γ s connection *coefficients* or Christoffel *symbols*? The answer is because, unlike most of the objects we deal with in relativity, they are not the components of a tensor.⁸

Example 7.5

Usually, we expect a tensor’s components to transform as

$$T^{\alpha'}_{\gamma'\beta'} = \frac{\partial x^{\alpha'}}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x^{\beta'}} \frac{\partial x^\gamma}{\partial x^{\gamma'}} T^\alpha_{\gamma\beta}. \quad (7.10)$$

We’ll see in Part V that the components of the connection actually transform as

$$\Gamma^{\alpha'}_{\gamma'\beta'} = \frac{\partial x^{\alpha'}}{\partial x^\alpha} \left(\frac{\partial^2 x^\alpha}{\partial x^{\beta'} \partial x^{\gamma'}} + \frac{\partial x^\beta}{\partial x^{\beta'}} \frac{\partial x^\gamma}{\partial x^{\gamma'}} \Gamma^\alpha_{\gamma\beta} \right). \quad (7.11)$$

This shows us that the first term in the braces spoils the tensor transformation law. Therefore, the Γ s are not tensors in general.

7.3 The covariant derivative

We now have the tools to finally take a covariant derivative. Substituting eqn 7.6 into eqn 7.5 we have a derivative

$$\frac{\partial \mathbf{v}}{\partial x^\alpha} = \frac{\partial v^\mu}{\partial x^\alpha} \mathbf{e}_\mu + v^\mu \Gamma_{\alpha\mu}^\lambda \mathbf{e}_\lambda, \quad (7.12)$$

and by relabelling the dummy indices

$$\frac{\partial \mathbf{v}}{\partial x^\alpha} = \left(\frac{\partial v^\mu}{\partial x^\alpha} + v^\lambda \Gamma_{\alpha\lambda}^\mu \right) \mathbf{e}_\mu. \quad (7.13)$$

From here on we shall write⁹ this quantity as $\nabla_\alpha \mathbf{v}$, which we call the covariant derivative of $\mathbf{v}$ along the direction $\mathbf{e}_\alpha$. This new notation allows us to replace the left-hand side of eqn 7.13 with $\nabla_\alpha \mathbf{v}$, rather than as $\partial \mathbf{v} / \partial x^\alpha$ which turns out to be a much more convenient.¹⁰ While we are in the process of introducing notation, a commonly used labour-saving shorthand for writing derivatives using commas and semicolons is given in the shaded box in the margin. Putting everything together, we have

$$\nabla_\alpha \mathbf{v} = \left(\frac{\partial v^\mu}{\partial x^\alpha} + v^\lambda \Gamma_{\alpha\lambda}^\mu \right) \mathbf{e}_\mu. \quad (7.18)$$

In terms of our classification of tensors, notice that $\nabla_\alpha \mathbf{v}$ is a (1,0) object, just like a vector.

Example 7.6

We can immediately note that in Minkowski spacetime the Cartesian basis vectors do not change with position and so all of the Γ coefficients are zero, giving the result that

$$\nabla_\alpha \mathbf{v} = \frac{\partial v^\mu}{\partial x^\alpha} \mathbf{e}_\mu \quad (\text{flat spacetime}). \quad (7.19)$$

We have worked out the covariant derivative along the direction of the basis vector $\mathbf{e}_\alpha$. What about the covariant derivative along an arbitrary vector? For this purpose we define the **connection operator** ∇ by writing $\mathbf{e}_\alpha \cdot \nabla = \nabla_\alpha$. We then define the action of our covariant derivative ∇_α on a scalar function simply as the derivative with respect to x^α , or

$$\mathbf{e}_\alpha \cdot \nabla f = \nabla_\alpha f = \frac{\partial f}{\partial x^\alpha}. \quad (7.20)$$

Generalizing to different directions by replacing $\mathbf{e}_\alpha$ with an arbitrary vector $\mathbf{u}$, we have the directional derivative

$$\nabla_{\mathbf{u}} f = \mathbf{u} \cdot \nabla f = u^\alpha \mathbf{e}_\alpha \cdot \nabla f = u^\alpha \frac{\partial f}{\partial x^\alpha}. \quad (7.21)$$

If we now interpret the action of the connection operator on vectors in the same way, $\nabla_{\mathbf{u}} \mathbf{v}$ is the covariant derivative of the vector $\mathbf{v}$ along the direction $\mathbf{u}$. We write this as

$$\nabla_{\mathbf{u}} \mathbf{v} = \mathbf{u} \cdot \nabla \mathbf{v} = u^\alpha \mathbf{e}_\alpha \cdot \nabla \mathbf{v} = u^\alpha \nabla_\alpha \mathbf{v}, \quad (7.22)$$

⁹We take ∇_α to be short for the more cumbersome

$$\nabla_\alpha \equiv \nabla_{\mathbf{e}_\alpha}. \quad (7.14)$$

In words, this is the directional derivative along the direction $\mathbf{e}_\alpha$.

¹⁰Relating the old notation to the new notation, we have

$$\frac{\partial \mathbf{v}}{\partial x^\alpha} \equiv (\nabla_\alpha \mathbf{v})^\mu \mathbf{e}_\mu. \quad (7.15)$$

Important here is that the components of the derivative are not necessarily equal to the derivatives of the components. This is due to the non-zero connection coefficients. In fact, we can rewrite our definition of the derivative of components and basis vectors in the new notation

$$\nabla_\alpha (v^\mu) = \frac{\partial v^\mu}{\partial x^\alpha}, \quad (7.16)$$

and

$$\nabla_\alpha (\mathbf{e}_\mu) = \Gamma_{\alpha\mu}^\lambda \mathbf{e}_\lambda. \quad (7.17)$$

Commas and semicolons:

Ordinary derivatives of functions can be written in **comma notation** as

$$\frac{\partial f}{\partial x^\alpha} \equiv f_{,\alpha}.$$

The covariant derivative is written in **semicolon notation** so that the μ component of eqn 7.18 becomes

$$(\nabla_\alpha \mathbf{v})^\mu \equiv v^\mu_{;\alpha} = v^\mu_{,\alpha} + v^\lambda \Gamma_{\alpha\lambda}^\mu,$$

and so $v^\mu_{;\alpha}$ are the components of the covariant derivative. This notation has the virtue of allowing some equations to be written more compactly, though this comes at the expense of leaving expressions littered with punctuation marks.

¹¹In semicolon notation

$$\nabla_{\mathbf{u}} \mathbf{v} = u^\alpha v^\mu{}_{;\alpha} \mathbf{e}_\mu,$$

where as before we have

$$v^\mu{}_{;\alpha} = v^\mu{}_{,\alpha} + v^\lambda \Gamma^\mu{}_{\alpha\lambda}.$$

and conclude that the covariant derivative along a vector $\mathbf{u}$ is given in terms of components by¹¹

$$\nabla_{\mathbf{u}} \mathbf{v} = u^\alpha \left(\frac{\partial v^\mu}{\partial x^\alpha} + v^\lambda \Gamma^\mu{}_{\alpha\lambda} \right) \mathbf{e}_\mu. \quad (7.23)$$

The covariant derivative selects out the change in a vector with position owing to its genuine change, eliminating the contribution due to the changing of coordinates with position. If a vector is parallel transported along a path then the only change should be due to the coordinates changing. So parallel transport of a vector $\mathbf{v}$ along the direction $\mathbf{u}$ implies

$$\nabla_{\mathbf{u}} \mathbf{v} = 0 \quad (\text{parallel transport}). \quad (7.24)$$

Example 7.7

This latter expression means that the components obey

$$u^\alpha \frac{\partial v^\mu}{\partial x^\alpha} + \Gamma^\mu{}_{\alpha\beta} v^\beta u^\alpha = 0, \quad (7.25)$$

or

$$\frac{\partial v^\mu}{\partial x^\alpha} = -\Gamma^\mu{}_{\alpha\beta} v^\beta. \quad (7.26)$$

In words, the change in the components $\partial v^\mu / \partial x^\alpha$ is, in this case, entirely due¹² to the change in coordinates $-\Gamma^\mu{}_{\alpha\beta} v^\beta$. That is to say that when a vector is parallel transported we have

$$\left(\begin{array}{c} \text{Change in a} \\ \text{vector's components} \end{array} \right) = \left(\begin{array}{c} \text{Change due to} \\ \text{coordinate system} \end{array} \right). \quad (7.27)$$

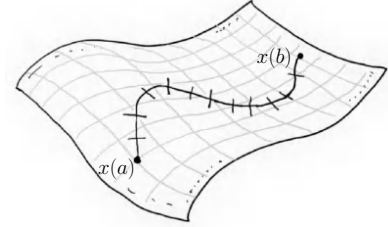


Fig. 7.4 The path $x(\lambda)$ is a curve through spacetime as a function of the parameter λ , stretching from the points associated with $\lambda = a$ to $\lambda = b$. The path marks off regular intervals, so we know how far along the curve we are.

7.4 Parametrized paths

We now have the covariant derivative at our disposal in the form of a directional derivative of some vector $\mathbf{v}$ taken along a vector $\mathbf{u}$. We shall also need the derivative in a form more suitable to apply to curves in spacetime such as the world lines of particles.

We saw in Chapter 1 that the most general way to describe a curve is to parametrize it by introducing a quantity that varies monotonically along its length. That is, a curve stretching from point $\lambda = a$ to $\lambda = b$ is written as $x(\lambda)$, where λ **parametrizes** the curve. It marks off regular intervals, so we know how far along the curve we are, as shown in Fig. 7.4.

Example 7.8

For the two-dimensional space $\mathbb{R}^2$ a curve is given by $(x(\lambda), y(\lambda))$, i.e. with x and y both functions of λ .

- A straight line $y = mx + c$ can be parametrized with $x(\lambda) = \lambda$ and $y(\lambda) = m\lambda + c$.
- A parabola $y = x^2$ can be parametrized with $x(\lambda) = \lambda$ and $y = \lambda^2$.
- A circle $x^2 + y^2 = a^2$ can be parametrized with $x(\lambda) = a \cos \lambda$ and $y(\lambda) = a \sin \lambda$.

The precise choice of parametrization isn't crucial. If an allowable parametrization is given by regular intervals of λ , we could equally well choose a different parametrization η such that $\lambda = \alpha\eta + \beta$, where α and β are constants. Such a parametrization is called an **affine parametrization**.

With this in mind, we can return to the covariant derivative itself. In many cases, we are interested in the rate of change of a **vector field**, $\mathbf{v}(x)$, an object where we input a position $x = \mathcal{P}$ and output a vector $\mathbf{v}$ appropriate for that point $\mathcal{P}$. We then ask how rapidly the vector field $\mathbf{v}(x)$ changes along a curve $x^\mu(\lambda)$. This involves parametrizing the curve, and then we seek

$$\left(\begin{array}{c} \text{Rate of change of} \\ \mathbf{v} \text{ with respect to } \lambda \end{array} \right) \equiv \frac{D\mathbf{v}}{d\lambda}. \quad (7.28)$$

Here we've introduced some new notation: the covariant derivative with respect to an affine parameter is denoted $D/d\lambda$.¹³

In order to use the covariant derivative as we've defined it so far, we seek a vector telling us the direction along which to take the derivative. This is provided by the tangent vector to the curve $x^\mu(\lambda)$, given by¹⁴

$$\mathbf{u} = \left(\frac{dx^\mu(\lambda)}{d\lambda} \right) \mathbf{e}_\mu. \quad (7.29)$$

That is, at every point λ along the curve we have a tangent vector $\mathbf{u}$ (Fig. 7.5). This tangent vector is one of the most useful tools in this book. We then have

$$\left(\begin{array}{c} \text{Rate of change of} \\ \mathbf{v} \text{ with respect to } \lambda \end{array} \right) \equiv \frac{D\mathbf{v}}{d\lambda} \equiv \nabla_{\mathbf{u}} \mathbf{v} \equiv \left(\begin{array}{c} \text{Covariant derivative of} \\ \mathbf{v} \text{ along } \mathbf{u} \end{array} \right), \quad (7.30)$$

where $\mathbf{u}$ is the tangent vector field to the curve. For massive particles, which follow timelike curves, we shall usually choose λ to be the proper time τ , which allows us to interpret the tangent as the particle's velocity, and provides the useful constraint $\mathbf{u} \cdot \mathbf{u} = -1$.

Example 7.9

This way of thinking about the covariant derivative makes it similar to an ordinary derivative defined in terms of evaluating a function at two points, $f(x)$ and $f(x+\delta x)$, and taking the difference in the limit of small δx . However, key to the definition here is the notion of parallelism, which allows us to remove the change caused by the changing coordinate system. In order to do this, the instructions of how to take the covariant derivative, in these terms, are as follows:

- (i) Take the vector $\mathbf{v}$ at $\lambda = \lambda_0 + \varepsilon$.
- (ii) Parallel transport it back to λ_0 .
- (iii) Evaluate $\delta\mathbf{v}$, which measured how different it is from $\mathbf{v}$ at λ_0 .
- (iv) Divide by ε and take the limit.

In equations, we have

$$\frac{D\mathbf{v}}{d\lambda} = \nabla_{\mathbf{u}} \mathbf{v} = \lim_{\varepsilon \rightarrow 0} \left(\frac{\mathbf{v}(\lambda_0 + \varepsilon)_{\text{(parallel transport to } \lambda_0)} - \mathbf{v}(\lambda_0)}{\varepsilon} \right). \quad (7.31)$$

This is illustrated in Fig. 7.6.

¹³The notation reminds us that owing to the changes in the coordinate systems, the components of the covariant derivative $(D\mathbf{v}/d\lambda)^\mu$, will not generally be equivalent to the derivatives of components $dv^\mu/d\lambda$. However, for a scalar field f we do have $df/d\lambda = Df/d\lambda$.

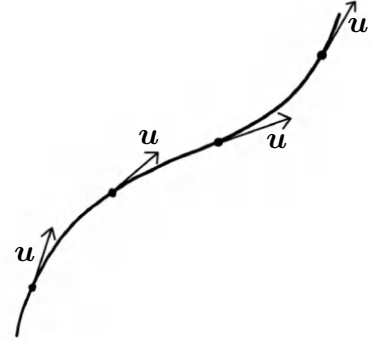


Fig. 7.5 The tangent vectors $\mathbf{u} = (dx^\mu(\lambda)/d\lambda) \mathbf{e}_\mu$ along the curve parametrized by λ . For $\lambda = \tau$ this provide a velocity vector (which itself varies along the path).

¹⁴We do not write $d\mathbf{x}/d\lambda$ as we sometime do in special relativity. As discussed in Chapter 3, the displacement vector $\mathbf{x} = x^\mu \mathbf{e}_\mu$, thought of as pointing a distance $|\mathbf{x}|$ from the origin to coordinate point x^μ , does not transform appropriately, and so we won't use it in this form (e.g. by taking its derivative). Note also that the tangent vector is given by $\mathbf{u} = (Dx^\mu(\lambda)/d\lambda) \mathbf{e}_\mu = (dx^\mu(\lambda)/d\lambda) \mathbf{e}_\mu$, since $x^\mu(\lambda)$ is a set of scalar functions.

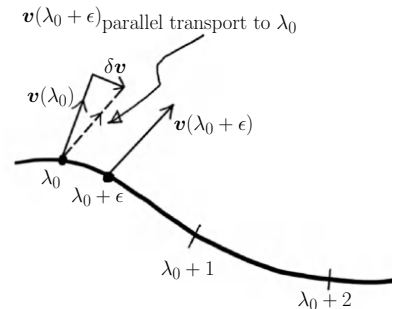


Fig. 7.6 Taking the covariant derivative using eqn 7.31.

Example 7.10

We can check our new version of the covariant derivative of a vector $\mathbf{A}$ in the case of flat spacetime, where the connection coefficients expressed in Cartesian coordinates vanish. Writing out all the components, we have

$$\begin{aligned} \left(\frac{D\mathbf{A}}{d\lambda} \right)^\alpha &= (\nabla_{\mathbf{u}} \mathbf{A})^\alpha = u^\mu \left(\frac{\partial A^\alpha}{\partial x^\mu} + \Gamma^\alpha_{\mu\nu} A^\nu \right) \\ &= u^\mu \frac{\partial A^\alpha}{\partial x^\mu} \quad (\text{the connection } \Gamma^\alpha_{\mu\nu} = 0) \\ &= \frac{dx^\mu}{d\lambda} \frac{\partial A^\alpha}{\partial x^\mu} = \frac{dA^\alpha}{d\lambda}, \end{aligned} \quad (7.32)$$

where we've used the chain rule in the final step. We conclude that for flat spacetime the components of the derivative are simply the derivatives of the components with respect to the parameter λ .

The covariant derivative notation $D/d\lambda$ proves very useful, not least because of its resemblance to the ordinary derivative.

7.5 Enter the metric

After formulating a covariant derivative, we might ask if this is the only way we could have constructed it. It turns out that our freedom to formulate it was restricted by the metric field $\mathbf{g}$, which is the foundation of our physical description of spacetime, and it is exactly this metric field that forces this version of the covariant derivative upon us. This idea is encapsulated in the notion of what is called the **compatibility of the connection** which requires that the covariant derivative obeys

$$\nabla_\alpha \mathbf{g} = 0 \quad \text{or, in components,} \quad g_{\mu\nu;\alpha} = 0, \quad (7.33)$$

for all α . This equation inseparably joins the metric and the covariant derivative.¹⁵ The importance of this condition is that, if it did not hold, then the lengths of vectors would change as we parallel transport them.¹⁶ This would be highly undesirable for a description of the physics of the real world.

Example 7.11

We shall prove the compatibility condition. The length of a vector is given by (the square root of) $\mathbf{A} \cdot \mathbf{A} = \mathbf{g}(\mathbf{A}, \mathbf{A})$, or $g_{\mu\nu} A^\mu A^\nu$. Take $\mathbf{A}$ to be a covariant constant (i.e. parallel) such that $\nabla_\alpha \mathbf{A} = 0$ (or $A^\mu_{;\alpha} = 0$). If the length of the vector is constant we expect the covariant derivative of $\mathbf{g}(\mathbf{A}, \mathbf{A})$ to vanish. Assuming the Leibniz product rule¹⁷ we have

$$g_{\mu\nu;\alpha} A^\mu A^\nu + g_{\mu\nu} A^\mu_{;\alpha} A^\nu + g_{\mu\nu} A^\mu A^\nu_{;\alpha} = 0. \quad (7.36)$$

Since $A^\mu_{;\alpha} = 0$, then we must have $g_{\mu\nu;\alpha} = 0$, as claimed.

¹⁵We do not yet have an explicit expression for how to compute the covariant derivative of a (0,2) tensor like $\mathbf{g}$. We delay deriving the explicit expression until Part V. At this stage we state that it is given in components by

$$g_{\mu\nu;\alpha} = g_{\mu\nu,\alpha} - \Gamma^\beta_{\alpha\mu} g_{\beta\nu} - \Gamma^\beta_{\alpha\nu} g_{\mu\beta}. \quad (7.34)$$

It is also helpful at this stage to note that we can also write the derivative for a (2,0) tensor like $\mathbf{T}$ in components as

$$T^{\mu\nu}_{;\alpha} = T^{\mu\nu}_{,\alpha} + \Gamma^\mu_{\alpha\beta} T^{\beta\nu} + \Gamma^\nu_{\alpha\beta} T^{\mu\beta}. \quad (7.35)$$

¹⁶Another consequence of this equation is that it provides the long-awaited explanation of what affine parametrizations actually are. They are those smooth parametrizations of a curve that have the property that the length of a vector doesn't change as we parallel transport the vector along the curve.

¹⁷The Leibniz product rule does indeed hold, as discussed in Part V.

Finally, we note that the compatibility condition is the basis of other links between the metric and the covariant derivative. The connection coefficients $\Gamma^\mu_{\alpha\beta}$ may be derived directly from the components of the metric.¹⁸ We saw how the connection coefficients arose due to the change in the basis vectors with position in spacetime and could be calculated via derivatives like $\partial \mathbf{e}_\mu / \partial x^\alpha = \Gamma^\lambda_{\mu\alpha} \mathbf{e}_\lambda$. Recall also that the components of the metric reflect the basis vectors via $g_{\mu\nu} = \mathbf{g}(\mathbf{e}_\mu, \mathbf{e}_\nu)$ or, more simply $g_{\mu\nu} = \mathbf{e}_\mu \cdot \mathbf{e}_\nu$. It therefore comes as little surprise that the connection coefficients are formed from a combination of first derivatives of the metric components as enshrined in the conceptual expression

$$\boxed{\partial g} \rightarrow \boxed{\Gamma}. \quad (7.38)$$

We shall expand on this point in the coming chapters.

In the next two chapters, we turn to the use of the covariant derivative in understanding how a particle moves under the influence of gravitation.

Chapter summary

- Parallel transport provides a method of comparing vectors at different points in curved space. A vector that is moved such that it has the same components in different local coordinate systems has been parallel transported.
- The covariant derivative can be used to measure how vector fields change with position in spacetime. It is a directional derivative given by

$$\nabla_{\mathbf{u}} \mathbf{v} = u^\alpha \left(\frac{\partial v^\mu}{\partial x^\alpha} + v^\lambda \Gamma^\mu_{\alpha\lambda} \right) \mathbf{e}_\mu. \quad (7.39)$$

- The tangent vector to the world line of a massive particle parametrized by the proper time is the timelike velocity vector $\mathbf{u}$, with the property $\mathbf{u} \cdot \mathbf{u} = -1$.

Exercises

(7.1) (a) Translate these expressions from component notation into index-free (i.e. tensor) notation

(i) $v^\alpha_{;\beta} u^\beta - u^\alpha_{;\beta} v^\beta$.

(ii) $u^\alpha v^\beta w_{\alpha;\beta}$.

If $\mathbf{T}$ is a symmetric (0, 2) tensor while $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are vectors, translate the following into component notation:

(iii) $\mathbf{v} \cdot (\nabla_{\mathbf{u}} \mathbf{T}) \cdot \mathbf{w}$.

(7.2) Consider a unit circle in the Euclidean plane. Parametrize the curve using an affine parameter s by writing $(x, y) = (\cos s, \sin s)$. Here s simply measures the distance on the curve, implying it must indeed be affine.

(a) Compute the components of the tangent vector to the curve $\mathbf{u}$, and its magnitude.

(b) Compute the components of $D\mathbf{u}/ds \equiv \nabla_{\mathbf{u}} \mathbf{u}$

¹⁸We shall see in Chapter 9 that the expression we will need is

$$g_{\rho\lambda} \Gamma^\rho_{\mu\sigma} = \frac{1}{2} \left(\frac{\partial g_{\lambda\mu}}{\partial x^\sigma} + \frac{\partial g_{\lambda\sigma}}{\partial x^\mu} - \frac{\partial g_{\mu\sigma}}{\partial x^\lambda} \right). \quad (7.37)$$

and verify $\mathbf{u} \cdot D\mathbf{u}/ds$ vanishes.

Now consider the reparametrization $t = \sin s$. This is not in the form $t = as + b$, so is not an affine parametrization.

(c) Recompute the components of the tangent vector $\mathbf{u}$, the derivative $D\mathbf{u}/dt$, and $\mathbf{u} \cdot D\mathbf{u}/dt$ using this new parametrization.

- (7.3) Consider the vector field $\mathbf{v}$ in two-dimensional flat space with Cartesian components $(v^x, v^y) = (0, Cx)$, with C a constant.

(a) Compute the vector $\nabla_\mu \mathbf{v}$ for $\mu = x$ and y .

(b) Convert the components of the vector into cylindrical polar coordinates using the transformations

from Chapter 3.

(c) Using the connection coefficients given in the chapter, compute the vectors $\nabla_\mu \mathbf{v}$ for $\mu = r$ and θ .

(d) Treat the quantity $(\nabla_\mu \mathbf{v})^\nu$ as components of a (1,1) tensor. Using the tensor transformation law, show that the results from part (c) are consistent with those of part (a).

- (7.4) The covariant derivative of a (0,2) tensor is written as

$$(\nabla_\alpha \xi)_{\mu\nu} = \frac{\partial \xi_{\mu\nu}}{\partial x^\alpha} - \Gamma^\lambda_{\alpha\mu} \xi_{\lambda\nu} - \Gamma^\lambda_{\alpha\nu} \xi_{\mu\lambda}. \quad (7.40)$$

Apply this to the components of the metric tensor and show, using eqn 7.37, that $(\nabla_\alpha g)_{\mu\nu} = 0$.